Model Evaluation, Explainability, and Fairness

A Business Analysis of Return Prediction for Retail E-Commerce Orders

Prepared by: Nhat Tam Huynh
Student ID: 101042941
Instructor: Zahra Atf
Course: MBAI 5310
Assignment 6
Dataset: Retail Product Return Dataset (370 orders, 16 columns)

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Introduction

This report evaluates a Decision Tree classification model that predicts whether a customer will request a return after placing a retail order. The analysis covers data understanding, preprocessing, model training, performance evaluation, error analysis, explainability (feature importance, SHAP, and LIME), and a fairness assessment across four customer subgroups: gender, age group, region, and membership level.

Each section below corresponds to one of the fifteen required analysis tasks, presenting the technical results alongside a business-oriented interpretation.

Task 1: Load and Understand the Dataset

The dataset contains 370 customer orders described by 16 columns, combining numerical order details with categorical customer and product attributes.

Dataset Composition

Category	Columns
Numerical features	customer_age, order_value, number_of_items, shipping_days, discount_percent, previous_returns, customer_rating
Categorical features	age_group, gender, region, membership_level, product_category, payment_method, delivery_issue_reported
Identifier	order_id
Target variable	return_requested

Data Quality

Check	Result
Dataset size	370 rows x 16 columns
Duplicate rows	0
Missing values: order_value	11 missing
Missing values: discount_percent	8 missing
All other columns	0 missing

Target Variable Distribution

Class	Count	Percentage
1: Return requested	259	70%
0: No return requested	111	30%

The dataset is imbalanced, with 70% of orders resulting in a return request. This imbalance is important to keep in mind for later evaluation: a model that simply predicted "return" for every order would already achieve 70% accuracy without learning anything useful.

Task 2: Define Features and Target Variable

Role	Columns
Target variable	return_requested
Model input features	customer_age, order_value, number_of_items, shipping_days, discount_percent, previous_returns, customer_rating, product_category, payment_method, delivery_issue_reported
Removed (identifier)	order_id
Reserved for fairness analysis only (not used as model input)	gender, age_group, region, membership_level

The four group-related columns (gender, age group, region, membership level) were deliberately excluded from model training. They are retained separately and reintroduced only in Task 14 to evaluate whether the model treats different customer groups fairly, without ever being used to make predictions directly.

Task 3: Data Preprocessing

Missing Value Treatment

Missing values in order_value (11 records) and discount_percent (8 records) were filled with the median of each column. After this step, the dataset contains zero missing values across all columns.

Categorical Encoding

All categorical input features (product_category, payment_method, delivery_issue_reported) were converted to numerical format using one-hot encoding (pd.get_dummies, with drop_first=True to avoid redundant columns).

Why Preprocessing Is Necessary

Machine learning models require numerical input and cannot directly process raw text values such as "Credit Card" or "Sports." Preprocessing also ensures missing values do not cause training errors or bias the model's learning. Both steps, imputing missing values and encoding categorical variables, are necessary before any model training can begin.

Task 4: Train/Test Split

Set	Records	Percentage
Training set	296	80%
Testing set	74	20%

The split used stratified sampling (stratify=y) and a fixed random_state=42 to keep the 70/30 return class balance consistent across both sets and to make the split reproducible.

Why We Split the Data

Training data teaches the model patterns; testing data simulates real-world conditions where the model encounters orders it has never seen. The model never sees the testing data during training, which is what makes the evaluation honest. Without this separation, there would be no way to know whether the model has learned generalizable patterns or simply memorized the training examples.

Task 5: Train a Classification Model

A Decision Tree Classifier (max_depth=3, random_state=42) was trained on the training set. The chosen depth of 3 was selected because it produces the best balance between training and testing accuracy, as confirmed later in Task 10.

What the Model Is Trying to Learn

The Decision Tree learns to ask a sequence of yes/no questions about each order, for example, "Did the customer have zero previous returns?" or "Is the customer rating below 3?", and follows a path through the tree to arrive at a prediction of Return or No Return.

Sample of Actual vs. Predicted Values (First 10 Test Records)

#	Actual Return	Predicted Return
0	1	1
1	0	0
2	1	1
3	0	0
4	1	1
5	1	1
6	1	1
7	1	1
8	1	1
9	0	1

Nine of the first ten predictions are correct. **Row 9** is a mistake: the actual value is 0 (no return) but the model predicted 1 (return), a **False Positive**. In a retail setting, this kind of error could lead to unnecessary proactive action such as pre-authorizing a refund or allocating warehouse space for a return that never arrives.

Task 6: Evaluate the Model

Metric	Score	Plain-Language Meaning
Accuracy	0.7432	74.3% of all 74 test orders were classified correctly.
Precision	0.7797	When the model predicted a return, it was correct 78.0% of the time.
Recall	0.8846	The model correctly identified 88.5% of all orders that were actually returned.
F1-Score	0.8289	The balanced harmonic mean of precision and recall.

Most Important Metric for This Business Problem

Recall is the most important metric for this business problem. The priority is to identify as many real return requests as possible so that warehouse, logistics, and customer service teams can prepare in advance. Missing a real return (a False Negative) is more disruptive than incorrectly flagging an order that will not be returned (a False Positive). A recall of 88.5% means the model catches the large majority of real returns before they happen.

Task 7: Confusion Matrix

	Predicted No Return (0)	Predicted Return (1)
Actual No Return (0)	9 (True Negative)	13 (False Positive)
Actual Return (1)	6 (False Negative)	46 (True Positive)

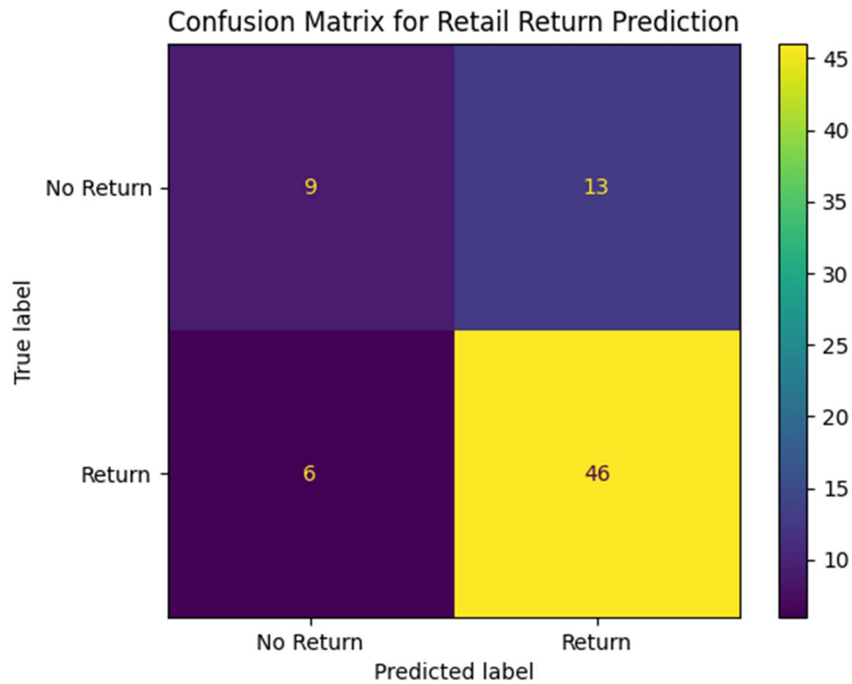

Figure 1. Confusion matrix for retail return prediction (test set, n = 74).

Business Meaning of Errors

- **False Positive (13 cases):** the company treats an order as a likely return when the customer has no intention of returning it, leading to unnecessary warehouse allocation, premature refund processing, or unwanted return instructions sent to a satisfied customer. The financial cost is relatively low but can affect customer experience.
- **False Negative (6 cases):** the company does not anticipate a return that is actually coming. This is more operationally disruptive; staff may be unprepared, inventory systems may not be updated in time, and the unexpected inflow can delay other operations.

Which Error Is More Serious?

False Negatives are more serious for this dataset. A return that arrives without being predicted can disrupt logistics planning, delay restocking, and create a poor customer experience if the return process is not ready. False Positives cause wasted preparatory effort but do not directly impact operations when the return does not arrive. Minimizing False Negatives, which corresponds to maximizing Recall, should be the primary business goal.

Task 8: Error Analysis

Error Type Counts

Error Type	Count
True Positive	46
False Positive	13
True Negative	9
False Negative	6

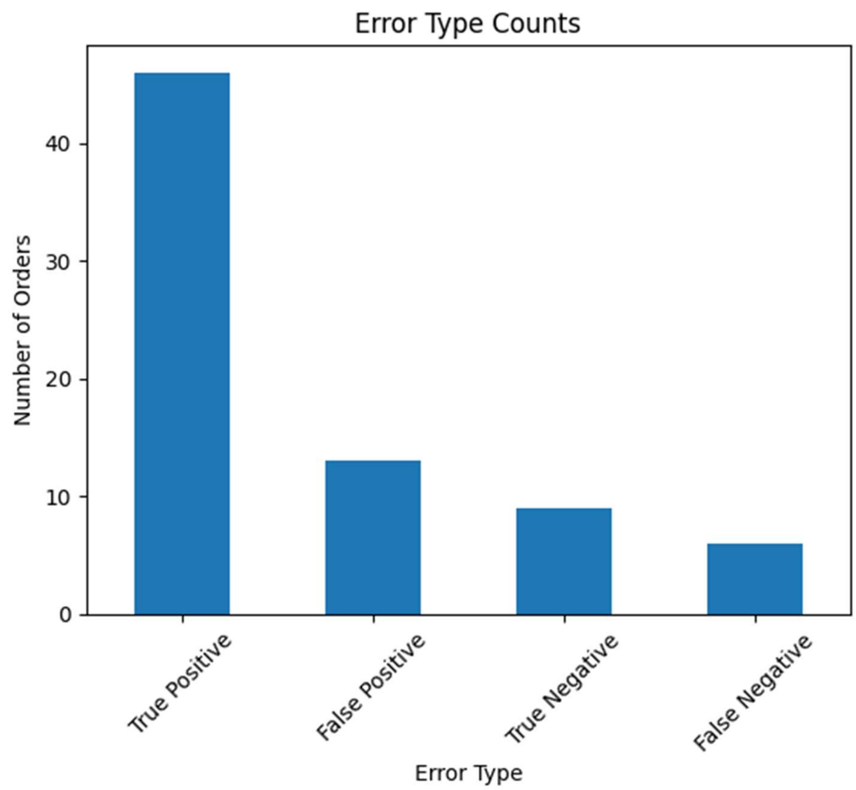

Figure 2. Count of each error type on the test set.

Sample of Wrong Predictions

Order ID	Actual Return	Predicted Return	Error Type
R0020	0	1	False Positive
R0056	0	1	False Positive
R0072	0	1	False Positive
R0055	1	0	False Negative
R0318	0	1	False Positive
R0046	1	0	False Negative
R0333	0	1	False Positive
R0159	0	1	False Positive

Order ID	Actual Return	Predicted Return	Error Type
R0141	1	0	False Negative
R0212	1	0	False Negative

Most Common Error Type and Business Impact

The most common error type is **False Positive (13 cases)**, where the model predicted would be returned, but was not. These are manageable from a business standpoint: wasted preparation, but no operational disruption. The less common but more damaging error is **False Negative (6 cases)**, unanticipated returns that arrive without warehouse or staffing readiness, which is the more serious operational risk discussed in Task 7.

Task 9: Cross-Validation

Fold	Accuracy
Fold 1	0.8649
Fold 2	0.6757
Fold 3	0.7432
Fold 4	0.7297
Fold 5	0.7568

Metric	Value
Mean Cross-Validation Accuracy	0.7541
Standard Deviation	0.0619

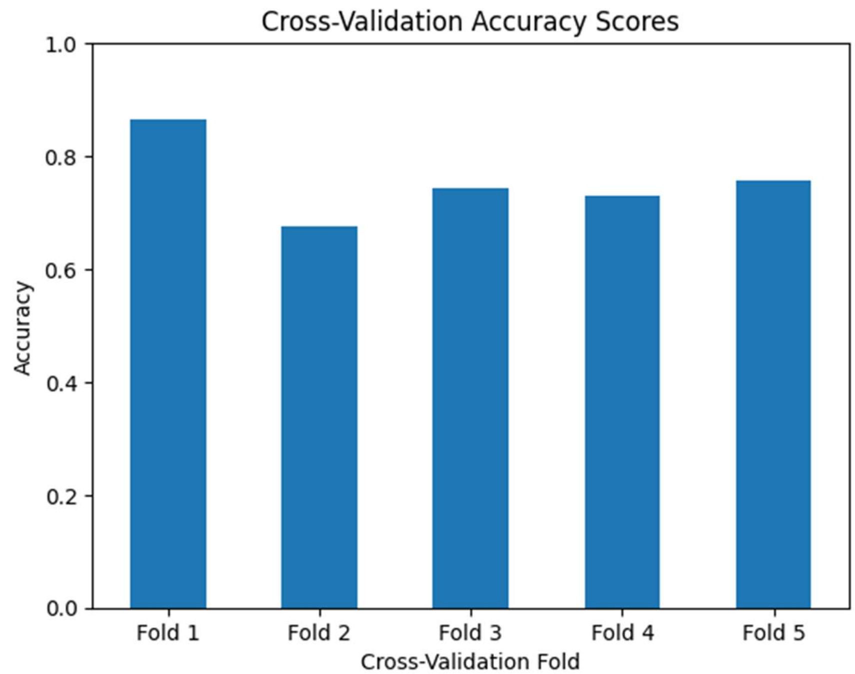

Figure 3. 5-fold stratified cross-validation accuracy.

Is Model Performance Stable Across Folds?

The model shows moderate stability across folds. Most folds cluster between 0.73 and 0.86, a reasonably consistent range. Fold 2 is the exception, dropping to 0.6757, likely because the imbalanced class distribution (70% returns) interacts differently across folds even with stratification. A mean accuracy of 75.4% is a meaningful improvement over the 70% baseline, and while the model is not perfectly stable, the variation is not extreme enough to call it unreliable.

Task 10: Overfitting and Underfitting Analysis

Dataset	Accuracy
Training Data	0.7905
Testing Data	0.7432

Accuracy Across Tree Depths

Max Depth	Training Accuracy	Testing Accuracy
1	0.6993	0.7027
2	0.7669	0.7297
3	0.7905	0.7432
4	0.8581	0.7027

Max Depth	Training Accuracy	Testing Accuracy
5	0.8919	0.7297
6	0.9324	0.6892
7	0.9662	0.6216
8	0.9899	0.6892
9	1.0000	0.6892
10	1.0000	0.6892

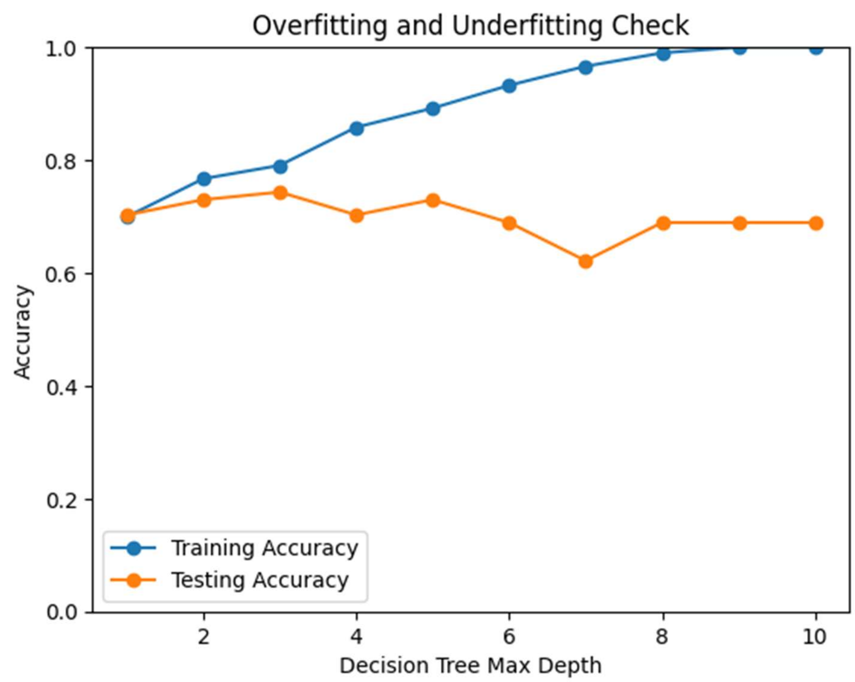

Figure 4. Training vs. testing accuracy across decision tree depths.

Overfitting / Underfitting Conclusion

- At **max_depth=3**, training accuracy is 0.7905 and testing accuracy is 0.7432, a small gap of about 4.7 percentage points, meaning the model generalizes well. This is why depth 3 was selected.
- Beyond depth 3, training accuracy climbs steadily, reaching 100% at depths 9–10, while testing accuracy peaks at depth 3 and then declines or flattens. At depth 9, the model achieves 100% training accuracy but only 68.9% test accuracy, a textbook case of **overfitting**: the model has memorized training records but fails to generalize.
- At the lowest depths (1–2), both training and testing accuracy are lower, showing mild **underfitting**, the model is too simple to capture the relevant patterns.
- **max_depth=3 is the sweet spot**, avoiding both overfitting and underfitting while keeping the training-testing gap small.

Task 11: Feature Importance

Feature	Importance
previous_returns	0.4104
customer_rating	0.2597
shipping_days	0.1819
discount_percent	0.1250
product_category_Home	0.0231
All other features (11 total)	0.0000

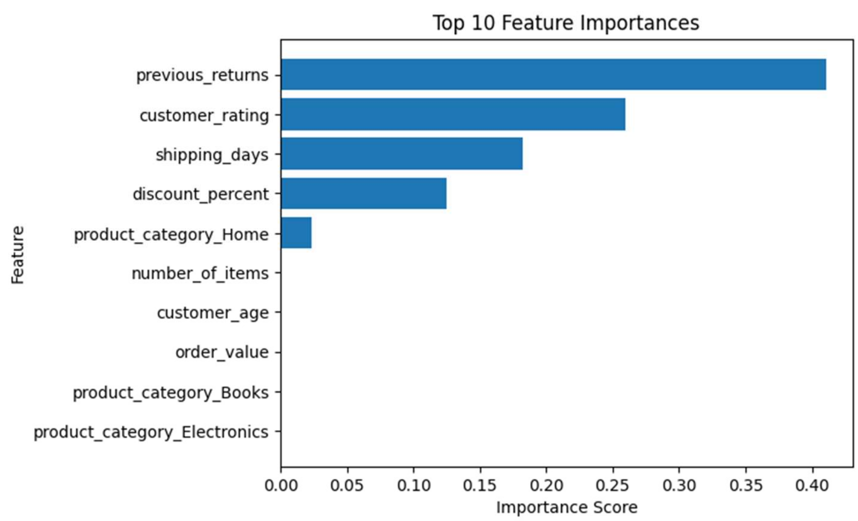

Figure 5. Top feature importances from the trained Decision Tree.

Business Meaning of the Top Features

- **previous_returns (0.41)**: the dominant variable. A customer who has returned products before is far more likely to return again; return behavior is habitual.
- **customer_rating (0.26)**: low satisfaction ratings correlate with returns; a low post-delivery rating is a strong warning signal.
- **shipping_days (0.18)**: longer shipping times correlate with higher return likelihood, possibly due to frustration or product condition concerns.
- **discount_percent (0.13)**: higher discounts correlate with higher returns, possibly reflecting impulsive purchases or lower commitment to keeping a heavily discounted item.

Note on Causation

Feature importance shows which features the model used when making decisions. It does not prove that these features directly cause returns. For example, it does not mean that giving a customer a larger discount will cause a return; the relationship may be indirect or driven by a third factor. Feature importance explains model behavior, not business causation.

Task 12: SHAP Explanation

SHAP (SHapley Additive exPlanations) was applied to the trained Decision Tree using shap.TreeExplainer. The base value, the average predicted probability of a return across the test set, is 0.6993.

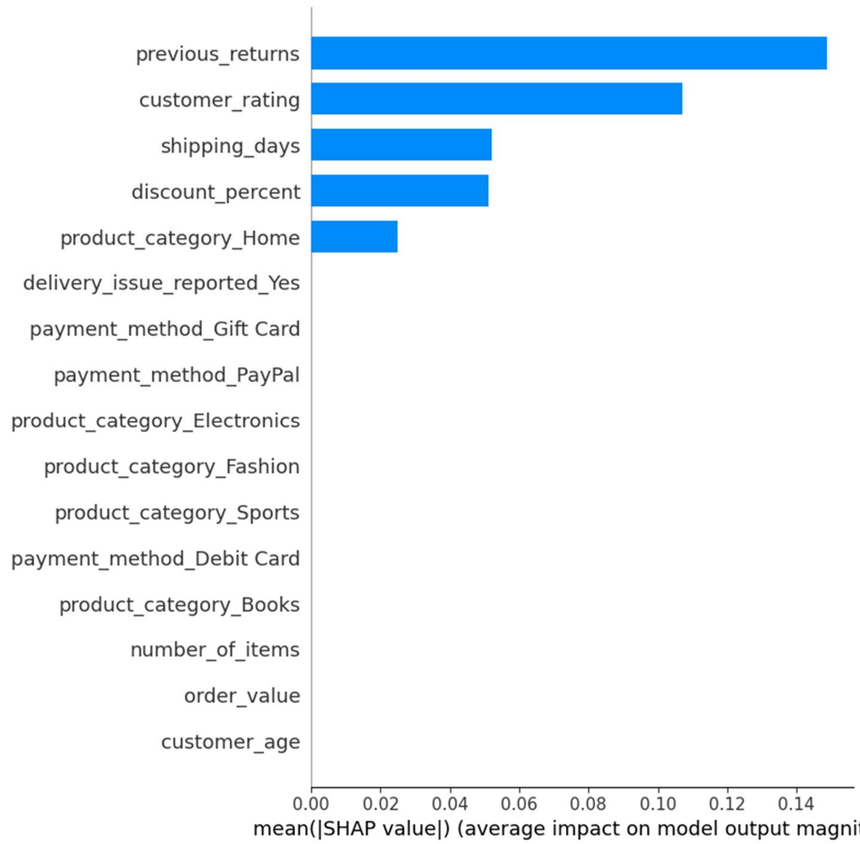

Figure 6. SHAP global feature importance summary (mean absolute SHAP value).

Individual Prediction: Order R0046

Feature	Value	SHAP Value	Direction
customer_rating	4	-0.2198	Pushes toward No Return
previous_returns	0	-0.1845	Pushes toward No Return
shipping_days	5	-0.0890	Pushes toward No Return
discount_percent	23.0	+0.0260	Pushes toward Return
product_category_Home	False	+0.0032	Pushes toward Return

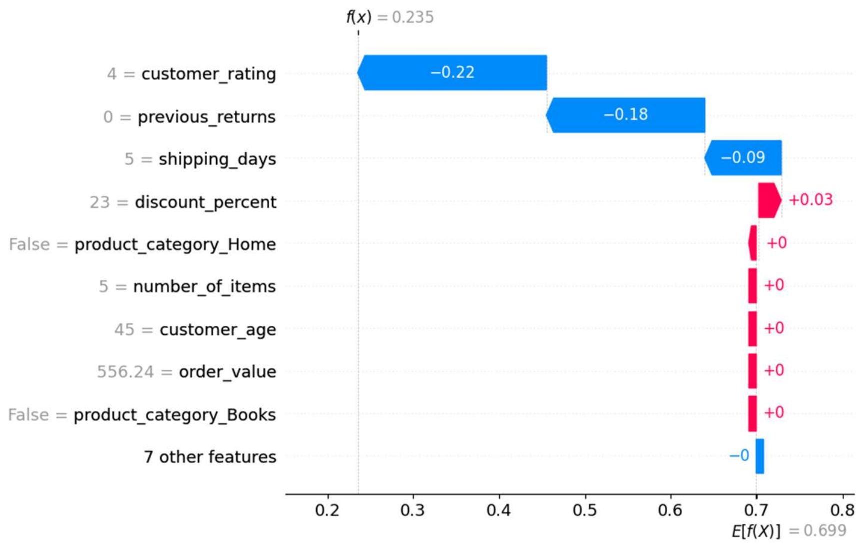

Figure 7. SHAP waterfall plot for Order R0046, showing how each feature shifted the prediction away from the base value.

How SHAP Helps Beyond Basic Feature Importance

Feature importance shows which features matter on average across all predictions. SHAP goes further by showing how much each feature pushed a specific prediction toward Return or No Return. SHAP explains both global model behavior and the reasoning behind individual predictions, making it more useful for detailed model understanding, and, as shown for Order R0046, for diagnosing exactly why a specific prediction went wrong.

Task 13: LIME Explanation

LIME (Local Interpretable Model-agnostic Explanations) was applied to the same record, Order R0046, to validate and complement the SHAP findings.

Actual Return	Predicted Return	Probability of No Return	Probability of Return
1	0	0.7647	0.2353

LIME Explanation Table

Feature Condition	Contribution
previous_returns <= 0.00	-0.3157 (toward No Return)
3.00 < customer_rating <= 4.25	-0.1487 (toward No Return)
product_category_Home <= 0.00	+0.0608 (toward Return)
22.50 < discount_percent <= 34.00	+0.0569 (toward Return)
4.00 < shipping_days <= 8.00	-0.0247 (toward No Return)

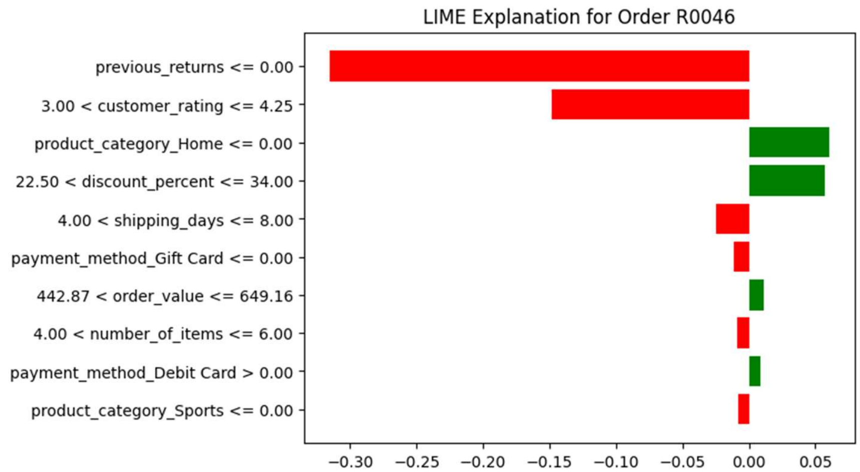

Figure 8. LIME explanation for Order R0046.

Interpretation: This Was an Actual False Negative

Order R0046 had **no prior return history** and a **rating of 4**, both of which the model treated as strong signals against a return. The model predicted No Return with 76.5% confidence, yet the customer did submit a return. This is exactly the kind of miss discussed in Task 7 and Task 8: a case where past behavior did not predict future behavior.

How LIME Helps Business Users

LIME translates the model's reasoning into plain conditions a business analyst or operations manager can read without ML knowledge. Instead of "the model predicted No Return because of weights in a decision tree," LIME says: "this order was predicted as No Return mainly because the customer has never returned anything before and gave a rating of 4." This makes the model's logic transparent and auditable. If a return arrives unexpectedly, the business can review the LIME explanation to see exactly what signals the model relied on, and what it missed.

Task 14: Fairness and Bias Reflection

Model performance was evaluated across four group-related columns, none of which were used as model inputs, to check whether the model treats different customer segments equally.

Fairness by Gender

Gender	Records	Actual Return Rate	Predicted Return Rate	Accuracy
Female	30	73.3%	76.7%	83.3%
Male	44	68.2%	81.8%	68.2%

The model is notably more accurate for Female customers (83.3%) than Male customers (68.2%), a 15-point gap. It over-predicts returns for male orders (81.8% predicted vs. 68.2% actual), while female predictions are much closer to actual outcomes (76.7% vs. 73.3%).

Fairness by Age Group

Age Group	Records	Actual Return Rate	Predicted Return Rate	Accuracy
18-25	10	80.0%	100.0%	80.0%
26-35	16	87.5%	81.3%	68.8%
36-50	23	52.2%	73.9%	69.6%
51+	25	72.0%	76.0%	80.0%

The most striking finding: for customers aged 18-25, the model predicted a return for every single one (100%), even though 20% did not actually return, an extreme over-prediction. The 26-35 and 36-50 groups have the lowest accuracy (68-70%); the 51+ group has the best accuracy (80%) and the closest match between actual and predicted rates.

Fairness by Region

Region	Records	Actual Return Rate	Predicted Return Rate	Accuracy
East	16	56.3%	81.3%	62.5%
North	16	75.0%	81.3%	81.3%
South	17	70.6%	88.2%	70.6%
West	25	76.0%	72.0%	80.0%

There is a large accuracy gap by region, from 62.5% (East) to 81.3% (North). In the East, actual returns are 56.3% but predicted returns are 81.3%, a severe over-prediction. The West region shows the best alignment between actual (76.0%) and predicted (72.0%) return rates.

Fairness by Membership Level

Membership	Records	Actual Return Rate	Predicted Return Rate	Accuracy
Basic	29	69.0%	72.4%	82.8%
Gold	15	53.3%	73.3%	53.3%
Platinum	8	75.0%	100.0%	75.0%
Silver	22	81.8%	86.4%	77.3%

Membership level reveals the most severe fairness concern. The model predicts a return for 100% of Platinum members, even though only 75% actually returned (small sample: 8 records). Gold members have the lowest accuracy (53.3%), close to random guessing, while Basic members show the best accuracy (82.8%).

Visual Comparison Across Groups

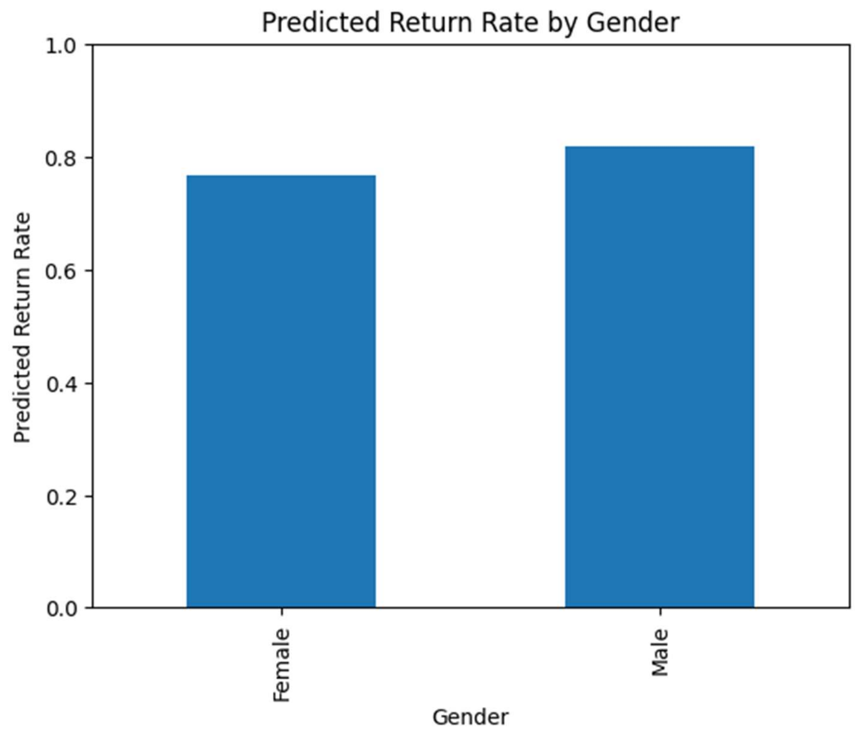

Figure 9. Predicted return rate by gender.

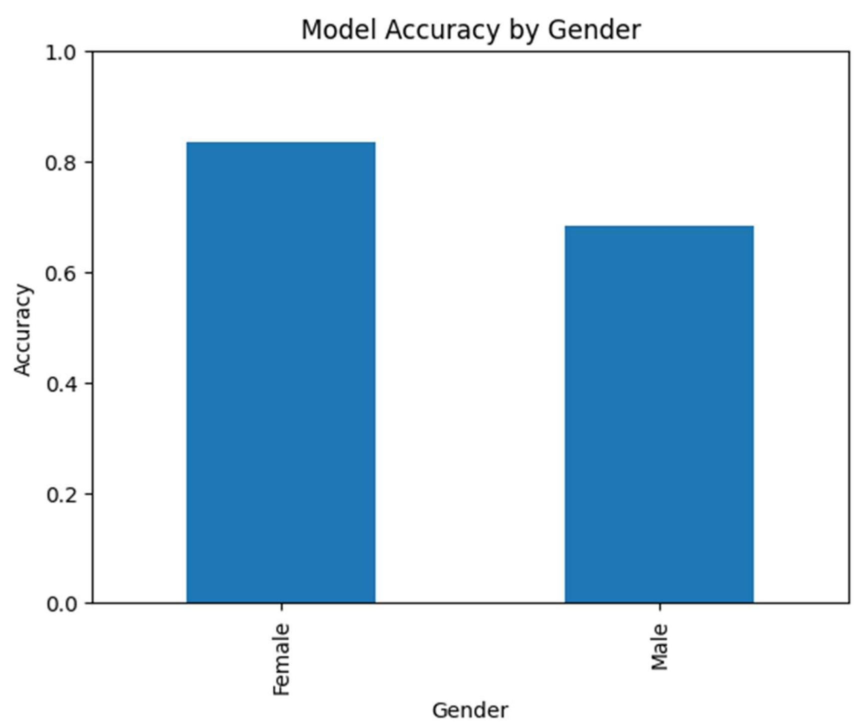

Figure 10. Model accuracy by gender.

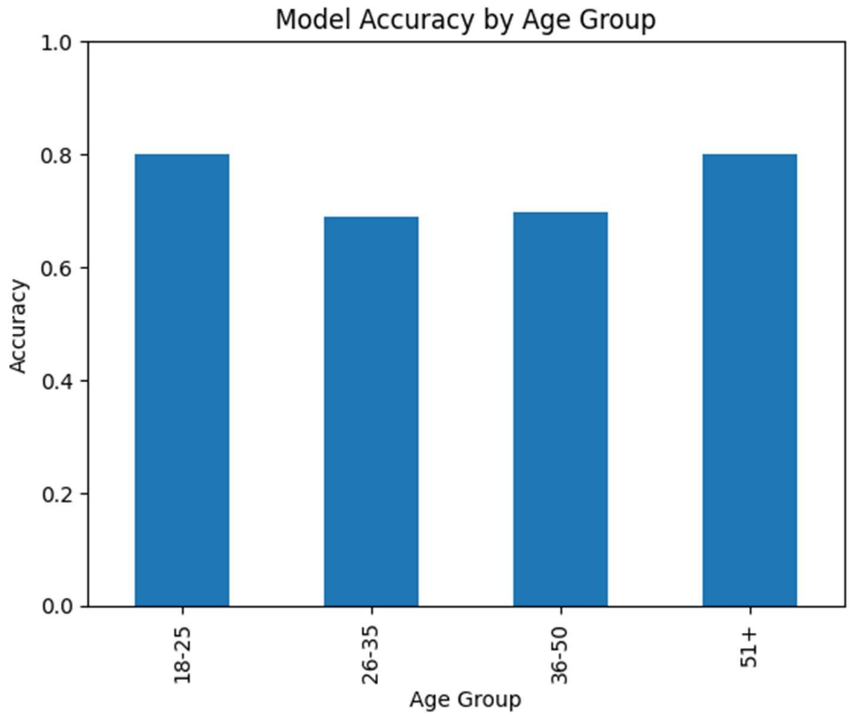

Figure 11. Model accuracy by age group.

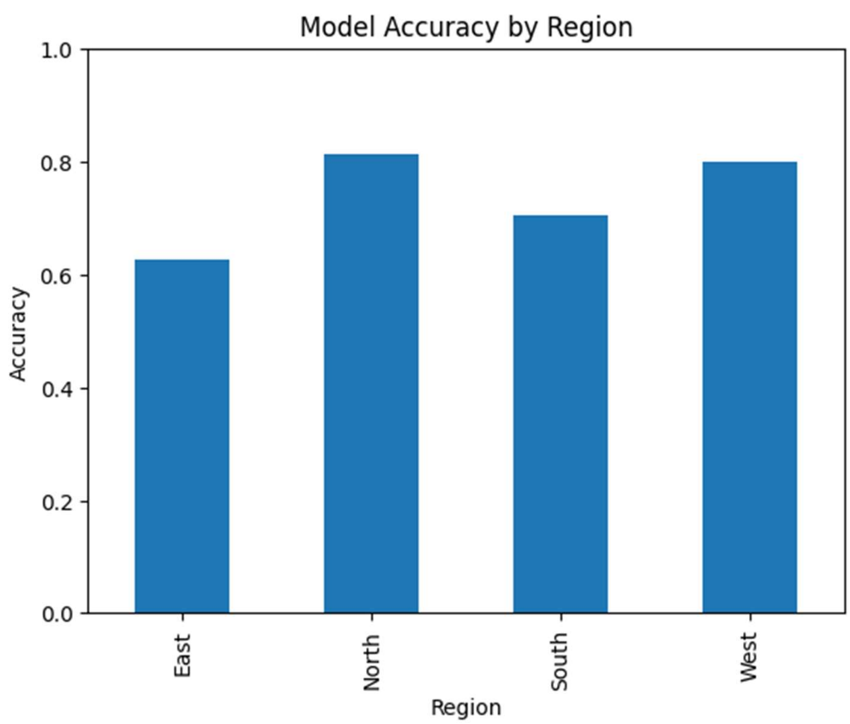

Figure 12. Model accuracy by region.

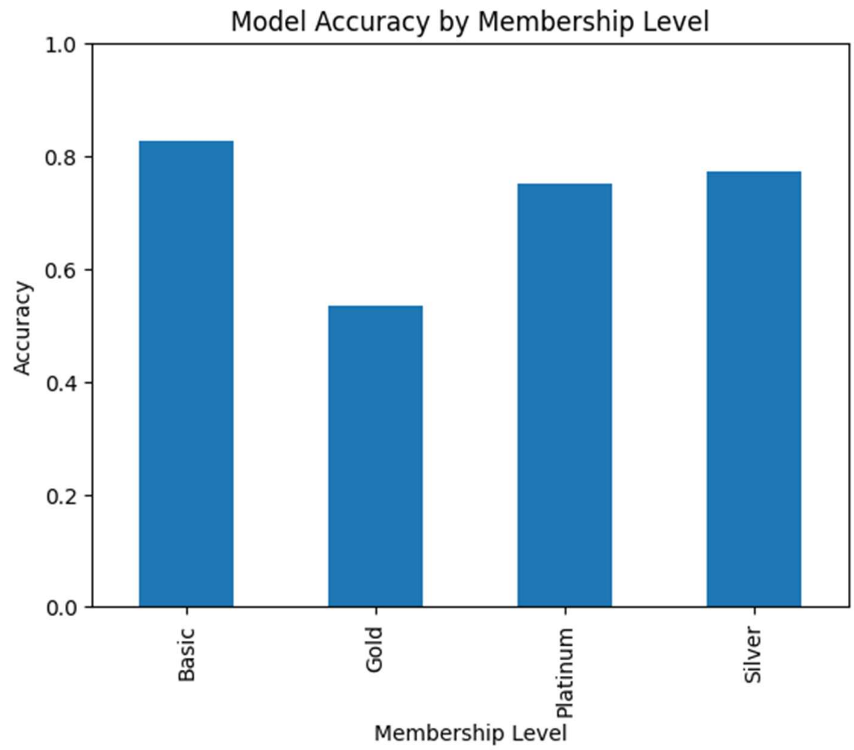

Figure 13. Model accuracy by membership level.

Overall Fairness Reflection

- The model does **not** perform equally across groups. Accuracy gaps of 15 points (gender), 18.8 points (region), and nearly 30 points (membership level, Basic vs. Gold) are large enough to create genuinely unequal business decisions.
- Even though none of these group columns were used directly as features, the model has effectively learned **proxies**. Features like previous_returns, customer_rating, shipping_days, and discount_percent are likely distributed differently across groups; for example, Platinum members may tend to have zero previous returns despite being high-value spenders, causing systematic over-prediction; the East region may have longer average shipping times, indirectly linking region to return risk.
- Before deploying this model for decisions that affect customer treatment, return approval workflows, loyalty offers, or shipping priority, the company should conduct a full fairness audit on a larger dataset and consider fairness-aware machine learning techniques to reduce group-level disparities.

Task 15: Final Business Interpretation

What the Model Predicts

This model predicts whether a customer will request a product return after placing an order on a retail e-commerce platform. It is trained on 296 orders and evaluated on 74 held-out orders. The target variable is binary: 1 means a return was requested, 0 means it was not.

Overall Model Performance

The model achieves a test accuracy of 74.3% using a Decision Tree with max_depth=3. This is above the 70% baseline that would result from predicting every order as a return. The mean cross-validation accuracy is 75.4% across 5 stratified folds, confirming that the test accuracy is a reliable estimate of real-world performance and not a result of a lucky data split. The training accuracy is 79.1%, giving a gap of roughly 4.7 percentage points, which is small and indicates the model generalizes reasonably well.

Most Important Evaluation Metric

For this business problem, Recall is the most critical metric. The company's primary concern is identifying real return requests in advance so that warehouse, logistics, and customer service operations can be prepared. A missed return (False Negative) is more disruptive than a false alarm (False Positive). The model achieved a recall of 88.5%, meaning it correctly identified 88.5% of all actual returns in the test set. This is the model's strongest result and the most relevant one for operational planning.

Main Error Type and Business Impact

The most common error is False Positive (13 of 19 total errors). The model flagged 13 orders as likely returns when the customers did not actually request a return. From a business perspective, this means warehouse space may be reserved, return packaging may be prepared, or customer service resources may be allocated for returns that never arrive. The cost is wasted preparation, not operational disruption.

The less common but more impactful error is False Negative (6 cases). Six orders resulted in actual returns that the model did not predict. These unanticipated returns create more serious disruption, as returns arrive without inventory or staffing being prepared to handle them. Given the business priority of recall, reducing these 6 missed predictions should be the primary improvement goal.

Overfitting and Underfitting Evidence

The model shows no meaningful signs of overfitting at max_depth=3. The 4.7 percentage point gap between training and testing accuracy is small. The overfitting analysis demonstrated clearly that increasing tree depth beyond 3 causes training accuracy to rise dramatically while test accuracy drops, reaching 100% training accuracy at depth 9 but only 68.9% test accuracy. The chosen depth of 3 represents the optimal balance for this dataset between learning enough patterns and avoiding memorization.

Most Important Features

The four features used by the model are previous_returns (importance 0.410), customer_rating (0.260), shipping_days (0.182), and discount_percent (0.125), with a minor contribution from product_category_Home (0.023). All other 11 features had zero importance in this tree.

The business interpretation of these features is meaningful: customers who have returned items before are the most likely to do so again; dissatisfied customers return products at higher rates; longer delivery times correlate with return requests, likely due to frustration or changed need; and large discounts attract purchases that customers may not keep. These four signals together account for all of the tree's predictive power.

SHAP and LIME Insight

The SHAP analysis confirmed the feature importance rankings and added directional understanding: high previous_returns pushes strongly toward predicting a return, while zero previous returns and a mid-to-high customer_rating push strongly away from return prediction. The LIME analysis on Order R0046, a False Negative case, illustrated a practical limitation of the model. The customer had no prior return history and gave a rating of 4, both of which the model treated as strong No Return signals. Yet the customer did submit a return. This suggests the model needs additional features, such as product-specific return rates, time since last purchase, or detailed delivery tracking data, to capture return risk in cases where behavioural history does not predict future behaviour.

Fairness and Bias Concern

The fairness analysis revealed meaningful disparities across all four group dimensions. The model is 15 percentage points more accurate for Female customers than Male customers. It predicts returns for 100% of 18-25 customers regardless of whether they actually returned. East region customers are over-flagged as returners, with only 62.5% accuracy. Gold membership customers receive the worst model accuracy at 53.3%, barely above random guessing. These disparities exist even though none of the group columns were used as direct model inputs, because the underlying features are correlated with group membership.

If this model were used to drive differentiated customer treatment, such as placing holds on certain accounts, routing return requests differently, or offering proactive incentives, these accuracy gaps would result in unequal service quality across demographic groups. This is a meaningful concern for responsible and ethical use of the model.

Final Recommendation

This model is suitable as an early prototype only and should not be used alone for final operational decisions.

Its recall of 88.5% is genuinely useful for anticipating the majority of returns, and the four key features it relies on, return history, satisfaction rating, shipping time, and discount level, are actionable and interpretable by business teams. However, several limitations prevent deployment as a standalone decision system:

- Overall test accuracy of 74.3% means roughly 1 in 4 predictions is wrong.
- Fairness gaps across gender, region, and membership level are large enough to create inequitable customer outcomes.
- The 6 False Negatives represent a meaningful operational gap in a high-volume retail environment.

Recommended next steps:

1. Collect more data to increase training size and reliability of minority-class predictions.
2. Engineer additional features, product-specific return rates, order fulfilment accuracy, and customer lifetime value.
3. Experiment with ensemble models (Random Forest, Gradient Boosting), which typically outperform single Decision Trees.
4. Apply fairness-aware training techniques to reduce group-level accuracy disparities.

With these improvements, the model could evolve from a prototype into a reliable operational tool.